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Ultrafast Carrier Dynamics in Individual Silicon Nanowires: Characterization of Diameter-Dependent Carrier Lifetime and Surface Recombination with Pump–Probe Microscopy

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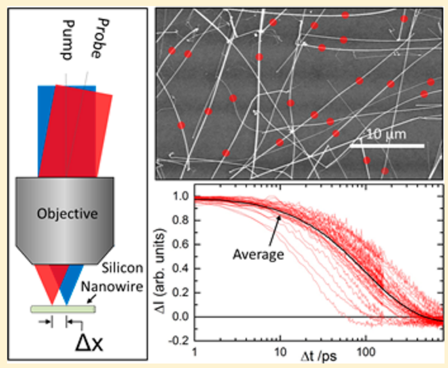
ABSTRACT: Ultrafast charge carrier dynamics in silicon nanowires (NWs) grown by a vapor–liquid–solid mechanism were interrogated with optical pump–probe microscopy. The high time and spatial resolutions achieved by the experiments provide insight into the charge carrier dynamics of single nanostructures. Individual NWs were excited by a femtosecond pump pulse focused to a diffraction-limited spot, producing photogenerated carriers (electrons and holes) in a localized region of the structure. Photoexcited carriers undergo both electron–hole recombination and diffusional migration away from the excitation spot on similar time scales. The evolution of the carrier population is monitored by a delayed probe pulse that is also focused to a diffraction-limited spot. When the pump and probe are spatially overlapped, the transient signal reflects both recombination and carrier migration. Diffusional motion is directly observed by spatially separating the pump and probe beams, enabling carriers to be generated in one location and detected in another. Quantitative analysis of the signals yields a statistical distribution of carrier lifetimes from a large number of individual NWs. On average, the lifetime was found to be linearly proportional to the diameter, consistent with a surface-mediated recombination mechanism. These results highlight the capability of pump–probe microscopy to quantitatively evaluate key recombination characteristics in semiconductor nanostructures, which are important for their implementation in nanotechnologies.



The optical and electronic properties of semiconductor nanomaterials are strongly influenced by their shape and composition, making them a promising platform on which to develop a variety of nanotechnologies.[1,2] While the strong structure–function relationship allows great flexibility in tuning these materials for specific device applications,[3−8] it also poses a great challenge for quantitatively characterizing and understanding dynamical phenomena. For example, surface recombination of photogenerated carriers is central to the performance of many active devices, and it becomes the dominant mechanism of carrier recombination as the sizes of these devices approach the nanoscale.[7,9,10] Direct measurement of the recombination rate can be obtained using common spectroscopic tools such as femtosecond transient absorption spectroscopy.[11,12] The difficulty with these spectroscopic methods is that they average over a large ensemble of nanostructures where there is frequently considerable variation in size and shape from one structure to the next. Furthermore, high-aspect-ratio materials (e.g., nanowires) often contain a distribution of secondary structures (i.e., local curvature in bent wires). The extensive averaging over the distribution of sizes, shapes, and secondary structures, as well as extraneous materials (residual reactants, catalysts, etc.) that may be present within the sample, makes it difficult to interpret transient signals obtained from ensembles in terms of fundamental dynamical processes. An alternative strategy involves performing measurements on single objects using time-resolved optical microscopy,[13−17] allowing the carrier dynamics to be correlated with shape, orientation, and morphology. Moreover, these methods can reveal spatially dependent phenomena, such as carrier diffusion and dynamical heterogeneity, at the single-structure level, which are inaccessible by conventional spectroscopic techniques.

In a previous paper, our group demonstrated the use of ultrafast microscopy to examine charge carrier recombination dynamics of silicon nanowires (NWs) on a wire-to-wire basis.[15] The high spatial resolution achieved by the optical microscope provides insight into the electron–hole recombination and carrier migration dynamics of a single nanostructure. In these experiments, an individual NW is excited by a femtosecond pump pulse that has been focused to a diffraction-limited spot by a microscope objective, producing photogenerated carriers (electrons and holes) in a localized region of the structure. After a well-defined delay, the excitation is probed by a second femtosecond laser pulse that is spatially overlapped with the pump pulse, and pump-induced changes to the transmission of the probe beam are detected. In this spatially overlapped pump–probe (SOPP) configuration, the decay of the signal reflects both electron–hole recombination and migration of carriers away from the excitation spot and out of the probe beam. Motion of the carriers is observed directly using a spatially separated pump–probe (SSPP) configuration, in

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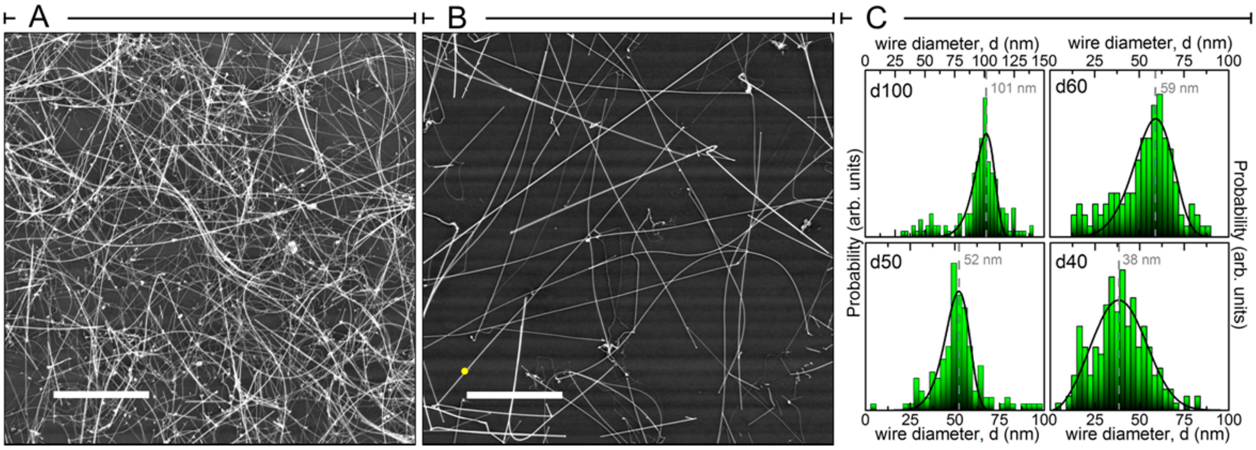

Figure 1. SEM characterization of NW samples. (A) SEM image of sample d50 in a region where NW density is high. TA measurements in ref 18 were performed in these higher density regions to maximize optical absorption. The scale bar is 10 μm. (B) SEM image of sample d50 in a region where NW density is lower. Microscopy measurements were performed in these lower density regions to isolate the spectroscopic signals from individual wires. The small yellow circle above the scale bar is 500 nm in diameter to illustrate the approximate size of the focused laser beams. The scale bar is 10 μm. (C) Diameter distributions for samples d100, d60, d50, and d40 measured by SEM (adapted from ref 18).

which carriers are created in one location and detected in another, thereby enabling the direct imaging of carrier diffusion.

In this paper, we expand on this previous work and characterize numerous wires to reveal a broad distribution of carrier lifetimes that is largely determined by wire-to-wire variations in diameter. Quantitative analysis of the data allows carrier diffusion along the length of the wire to be decoupled from the measured kinetics and shows that carrier recombination is consistent with a surface-mediated mechanism in which the lifetime (τ) is proportional to the NW diameter (d), i.e., $\tau = d/4S$. The surface recombination velocity (S) provides a measure of surface quality and has both fundamental and technological significance. It offers a quantitative means of studying the interaction of carriers with the surface and for comparing different surface passivation strategies. It can also be related to the density of surface states, which governs the behavior of many active electronic components, making it of interest to many nanotechnologies. Pump−probe microscopy enables the measurement of carrier lifetimes at specific points within an individual structure, which can then be used, with the nanowire diameter, to extract a value for S. We compare the microscopy results to those obtained on the same set of samples using femtosecond transient absorption (TA) spectroscopy, which measures the ensemble-averaged lifetimes.[18] The TA-measured lifetimes are 2−3 times shorter than those observed by microscopy, a discrepancy that may arise from differences in the distribution of secondary structures accessed in the two experiments. The combined set of results demonstrates that the two methods, ensemble TA and pump−probe microscopy, provide complementary views of the carrier dynamics in semiconductor nanostructures.

■ EXPERIMENTAL SECTION

Intrinsic Si NWs were grown by a vapor−liquid−solid mechanism[1] using a home-built, hot-wall chemical vapor deposition (CVD) system.[6] For a typical growth run, Au nanoparticles with diameters of ∼30, 50, 60, and 100 nm were dispersed on quartz substrates. NWs were grown with a total reactor pressure of 40 Torr using a gas flow of 2.00 standard cubic centimeters per minute (sccm) of silane and 200 sccm of hydrogen as carrier gas. The reactor was held at 450 °C for 2 min to nucleate wire growth and then cooled (12 °C/min) to 410 °C for continued wire growth over 2 h. After completion of wire growth, NWs were thermally oxidized at 1000 °C for 60 s in 100 Torr of flowing oxygen to form a 5−10 nm thick thermal oxide. The samples were then immersed in 2-propanol, which upon evaporation collapses the NWs onto the plane of the quartz substrate, forming the dense mat of randomly oriented, overlapping NWs shown in Figure 1A. Scanning electron microscopy (SEM) images reveal a large number of high-quality NWs that remain largely unbroken upon collapse (Figure 1A).

The pump−probe microscope has been described in detail elsewhere.[15] Briefly, a portion of the output beam comprised of 100 fs fwhm (full width at half-maximum) pulses centered at 850 nm is split, and each arm is passed through a pair of synchronized acousto-optic modulators (AOMs; Gooch and Housego), which reduce the pulse repetition rate to 1.6 MHz. The pump beam is frequency doubled by a β-barium borate (BBO) crystal, attenuated to 2.5 pJ/pulse, and modulated at a 50% duty cycle by the AOM. It was verified that further reduction of the pump power did not change the transient kinetics. The probe beam (630 fJ/pulse) is passed onto a mechanical delay stage and a pair of computer-controlled mirrors before being collinearly coupled onto the pump beam path with a dichroic mirror. Despite the comparable fluence of pump and probe beams, the probe beam only weakly interacts with the Si NW sample due to a 100-fold lower extinction coefficient at 850 nm compared to 425 nm.[19] Pump and probe beams are focused onto the sample with a 0.8 NA 100× ultralong working distance objective. The transmitted probe beam is collected with a condenser, passed through a filter to attenuate transmitted pump light, and detected on one channel of a balanced photodetector. The other channel is balanced with a portion of the probe beam picked off before the microscope. The change in transmittance is detected by lock-in detection (SR810, Stanford Research) and processed with home-written Matlab software. All measurements were performed at room temperature in an ambient atmosphere.

RESULTS AND DISCUSSION

Nanowire Samples. Microscopy measurements were performed at the edges of the densest regions so that individual NWs could be resolved without multiple overlapping wires contributing to the measured signal (Figure 1B). These experiments focus on four samples denoted d40, d50, d60, and d100 with average diameters of 38, 52, 59, and 101 nm, respectively (Figure 1C).[18]

Pump–Probe Microscopy. Carrier recombination dynamics were monitored with a pump–probe microscope that possesses femtosecond temporal resolution and near-diffraction-limited spatial resolution. The microscope, which uses an objective to focus the 425 nm pump and 850 nm probe beams to near-diffraction-limited spots within a single structure, can be operated in two modes, SOPP and SSPP, as discussed above. A comparison of conventional bright-field and SOPP images for the d50 sample is displayed in Figure 2. The bright-field image (Figure 2A) shows a number of individual NWs in the field of view. The pump–probe images (Figure 2B,C), which map the spatial variation in the transient absorption signal,[20,21] were obtained by holding the delay fixed at $\Delta t = 0$ ps and raster scanning the sample underneath the spatially overlapped pulses. At early times, photoexcitation by the pump pulse causes an increase in the detected intensity of the probe beam, i.e., $\Delta I = I_{\text{pump on}} - I_{\text{pump off}} > 0$, which is represented by warmer colors in the color map. This photoinduced transparency is partially attributed to changes in NW absorption resulting from band filling by photoexcited free carriers that occupy low-energy states in the conduction and valence bands of Si. The intensity of the probe beam is also affected by changes in NW scattering arising from shifts in the index of refraction brought on by photoexcitation. While it is difficult to disentangle the relative contributions of absorption and scattering, in either case the associated increase in probe transmittance (the bleach signal) is proportional to the free carrier density, and thus provides a means to monitor changes in the electron and hole populations. In addition to this positive-going bleach, photoinduced absorption of free carriers and trapped carriers may also contribute to the overall transient spectra as a negative-going signal. However, consistent with previous studies of silicon, the overall signal is positive at early times, suggesting that carrier absorption (free or trapped) is a minor component relative to band filling.[11,16,22,23]

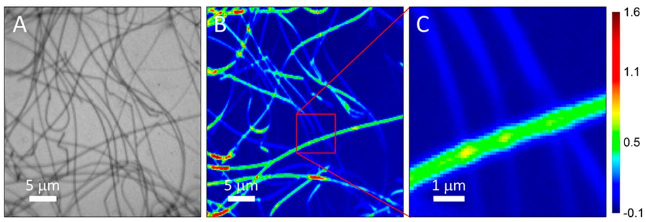

Figure 2. Comparison of conventional bright-field and transient SOPP imaging of NW samples. (A) Conventional bright-field optical image of a ∼40 μm × 40 μm region of the d50 sample containing 50 nm diameter NWs. (B) SOPP image of the sample region shown in panel A and collected at a pump–probe delay of $\Delta t = 0$ ps. (C) Higher resolution image of the boxed region in panel B. The spatial resolution of SOPP images is ∼500 nm fwhm and is comparable to the resolution of the bright-field image.

The variation in signal magnitude observed between and within individual NWs across the SOPP image is not the result of spatial heterogeneity in the intrinsic properties of the NW (e.g., crystallinity, defect density, surface passivation), but rather stems from a combination of extrinsic factors. For example, overlapping wires result in greater signal size compared with other NW sections. NWs located at the focal plane experience the highest pump intensity, generating a greater carrier density and larger transient signals. There is also a polarization effect. The images in Figure 2B,C were collected with the pump and probe polarization vectors directed along the horizontal axis of the image. While the decay kinetics are polarization independent, the magnitude of the pump–probe signal is larger when the probe polarization is directed parallel to the NW axis. Thus, NWs with significant curvature exhibit substantial variation along the growth axis, with segments aligned along the probe polarization appearing brighter. Indeed, SOPP images obtained from straight isolated NWs show very little variation in signal intensity along the wire axis, indicative of the production of highly uniform structures by the VLS growth process. Thus, the spatial variation observed in the collapsed NW films is most likely a reflection of extrinsic experimental factors and does not indicate a lack of material uniformity.

Distribution of Decay Kinetics. The decay kinetics of 25–50 individual NWs were collected with spatially overlapped pump and probe beams (SOPP configuration) in each of the four samples. The kinetic traces measured from each point are shown in blue in Figure 3 and reveal that a single sample possesses a wide distribution of decay rates. While there is significant variation from one trace to the next, they share a common decay profile in which the transient bleach (positive ΔI) decays over several hundred picoseconds as result of electron–hole recombination, eventually changing sign to yield a low-amplitude absorptive component that persists beyond 1 ns. The negative-going component has characteristics that are consistent with a combination of thermal effects resulting from localized heating by the pump pulse[24] and trap carrier absorption.

To facilitate a direct comparison with TA results, the 25–50 kinetic traces from each sample were averaged, producing the single kinetic trace shown in black in each panel of Figure 3. A comparable decay curve collected by TA at $\lambda_{\text{probe}} = 650$ nm is depicted by the gray squares.[18] While qualitatively similar, the average decay collected with SOPP microscopy (black curve) is generally slower than the corresponding TA measurement (gray squares). This discrepancy is not caused by probing different regions of the spectrum as the photoinduced bleach is a spectrally broad feature that shows identical decay kinetics at $\lambda_{\text{probe}} = 650$ nm and $\lambda_{\text{probe}} = 850$ nm.[18] Rather, as we will discuss in more detail below, the shorter TA-measured lifetimes likely stem from greater contributions from both bent NWs and non-NW structures.

Contribution of Charge Carrier Diffusion. The decay of the transient absorption signal obtained in the SOPP measurement reflects the loss of free carrier population in the probe region resulting from electron–hole recombination and diffusional migration of charge carriers out of the excitation volume. Motion of the charge carriers is directly observed in SSPP images (Figure 4B). In this configuration, the pump–probe delay time is fixed and the probe beam is raster scanned with respect to the stationary pump beam by means of a scanning mirror assembly. The photoinduced transmission change is measured on a pixel-by-pixel basis, resulting in an image of the spatial distribution of carriers.

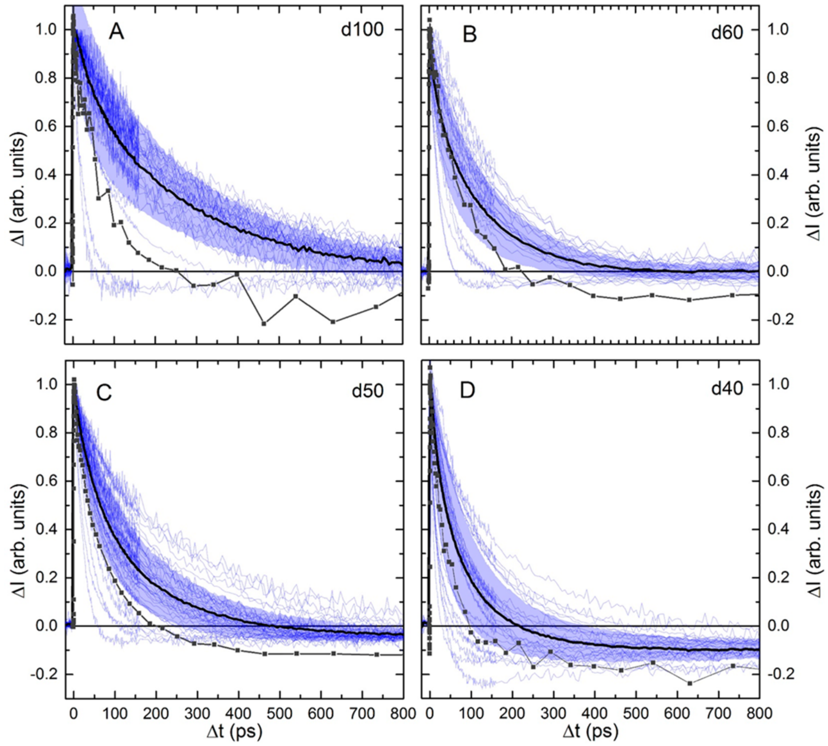

Figure 3. Comparison of diameter-dependent recombination kinetics from SOPP microscopy and TA spectroscopy for samples (A) d100, (B) d60, (C) d50, and (D) d40. SOPP measurements acquired at multiple spatial positions within each sample are shown in light blue. The average and standard deviation of these SOPP measurements are shown as a solid black curve and shaded blue region, respectively. Transient kinetics measured with TA at λ_{probe} = 650 nm are shown as gray squares.

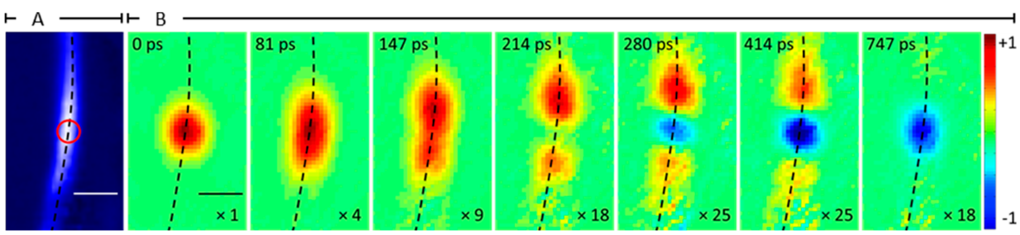

Figure 4. Imaging of charge carrier diffusion. (A) SOPP image of an NW within the d50 sample. The scale bar is 1 μm. The orange circle indicates the position of the pump beam for measurements in panel B. (B) SSPP images collected at the delay time denoted in the upper left corner of each image. The scale bar is 1 μm. The dashed black lines are visual guides indicating the location of the NW. The line thickness reflects the average diameter of the sample (52 nm). Each image is depicted using a normalized color table with the normalization factor denoted in the lower right corner of each image.

The SSPP image at $\Delta t = 0$ ps shows a circular, positive-going signal that reflects the photoinduced transparency resulting from the localized distribution of free carriers produced by the pump pulse. This image is well described as a two-dimensional Gaussian with an fwhm of ~700 nm, providing a measure of the lateral resolution of the SSPP method.[25] At longer time delays, the carrier cloud elongates along the NW axis, providing a direct observation of the free-carrier diffusional motion away from the excitation spot. As the free carriers recombine and diffuse away from the excitation spot, a low-amplitude absorptive component (blue, negative signal in Figure 4B) emerges that, unlike the free carrier signal, remains largely localized at the point of excitation. This static size is consistent with the assignment of the negative-going signal to a combination of trap carrier absorption and localized heating,[18] both of which should exhibit little spatial expansion compared to the free carriers. Trapped carriers move slower than free carriers because they are confined to localized states within the lattice, and thermal diffusion in silicon ($\alpha = 0.8$ cm^2/s) is more than an order of magnitude slower than the ambipolar carrier diffusion constant ($D_B = 18$ cm^2/s).[26]

While the SSPP images in Figure 4B provide a valuable visualization of the carrier motion, acquisition of images at a sufficient number of time delays to permit a quantitative kinetic analysis is not experimentally feasible. Instead, we have monitored the decay of the transient absorption signal as a function of time with the spatial separation (Δx) between the pump and probe spots along the NW held fixed, as shown in Figure 5. Panel A shows an SOPP image of an NW from sample d50 at $\Delta t = 0$ ps. The four different kinetic traces in Figure 5B are produced by positioning the pump beam at the location marked by the arrow in Figure 5A with the probe beam located at the black, red, green, and blue circles. The decay obtained with $\Delta x = 0$, which is identical to that of the SOPP configuration described above, exhibits its maximum signal at $\Delta t = 0$. When the probe is separated from the pump (i.e., Δx >

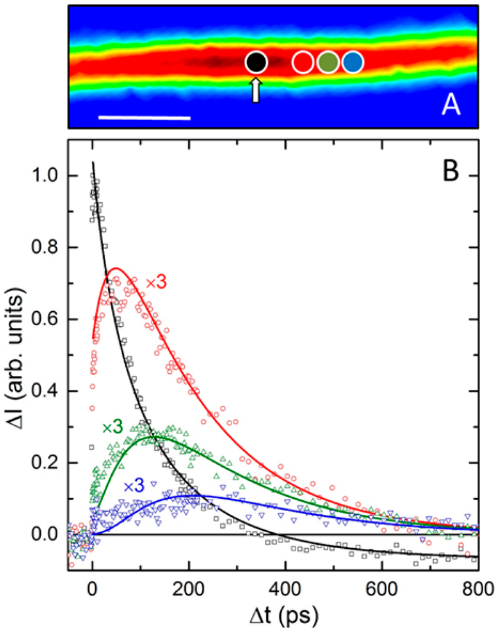

Figure 5. Kinetics of charge carrier diffusion. (A) SOPP image of a single NW in the d50 sample collected at a pump−probe delay of Δt = 0 ps. The wire is excited at the location indicated by the arrow, and colored circles denote the position of the probe beam for the kinetics displayed in panel B. The scale bar is 1 μm. (B) SSPP transient kinetics collected at pump−probe separations Δx = 0.0 (black squares), 0.56 (red circles), 0.87 (green triangles), and 1.18 (blue inverted triangles) μm indicated by the corresponding colored circles in panel A. Solid lines show a global fit of the experimental data to eq 3.

0), the maximum free carrier signal occurs at Δt > 0 ps, reflecting the time necessary for the photoexcited carriers to diffuse from the excitation region and arrive at the probe location.

The carrier diffusion and electron−hole recombination contributions to the transient absorption signal in the SSPP experiment, $I_{\Delta x}(t)$, are described by

$$I_{\Delta x}(t) = A_{\Delta x}(t) \exp(-t/\tau_r) + a_\infty \tag{1}$$

where τ_r is the carrier lifetime, Δx is the displacement between the pump and probe, and a_∞ is an offset to account for the long-lived absorptive component. The impact of carrier motion on the transient absorption signal is contained in the time-dependent pre-exponential factor, $A_{\Delta x}(t)$, which is given by the double convolution over the spatial coordinates of pump and probe beams with the linear diffusion equation. Assuming Gaussian beam profiles, this function can be expressed as

$$A_{\Delta x}(t) = \frac{a_0}{\gamma_2\gamma_2\sqrt{D_B t}} \int_{\infty}^{-\infty} dx' \exp\left(\frac{-4(x' - \Delta x)^2 \ln(2)}{\gamma_2^{\ 2}}\right) \int_{\infty}^{-\infty} dx'' \exp\left(\frac{-4(x'')^2 \ln(2)}{\gamma_1^{\ 2}}\right) \exp\left(\frac{-(x' - x'')^2}{4\pi D_B t}\right) \tag{2}$$

where γ_1 and γ_2 are the pump and probe spatial fwhms, respectively, D_B is the ambipolar bulk diffusion constant for silicon (18 cm^2/s), and a_0 is a time-independent intensity scaling factor. The integrals in eq 2 can be solved analytically and substituted into eq 1 to give

$$I_{\Delta x}(t) = \frac{a_0}{\beta(t)} \exp\left(\frac{(-4\ln(2)\Delta x^2)}{\beta(t)^2}\right) \exp(-t/\tau_r) + a_\infty \tag{3}$$

where $\beta(t) = (\gamma_1^{\ 2} + \gamma_2^{\ 2} + 16 D_B t \ln(2))^{1/2}$. Implicit in this expression is the assumption that both pump and probe absorption events are linear; that is, the photogenerated carrier population is proportional to the intensity of the pump beam, and the change in probe intensity is proportional to the pump-induced carrier population. In ref 15, we demonstrated qualitative agreement between the model and observed carrier dynamics in Si NWs. The analytical form expressed by eq 3 allows a quantitative evaluation of both diffusion and recombination dynamics of the photogenerated carriers.

The solid lines in Figure 5B show results from a global fit of the four transients to eq 3. The pump and probe spot sizes (γ_1 = 310 nm and γ_2 = 620) were determined by deconvolution of the Δt = 0 ps SSPP image (first panel in Figure 4B) assuming the pump is half the diameter of the probe. The offset used to account for the long-lived absorptive contribution was fixed at a_∞ = −0.68 in the Δx = 0 data but set to zero for the others. All four kinetic traces are well fit by the model using only the shared parameters, a_0 and τ_r, suggesting the relevant diffusion and carrier recombination processes are adequately described by eq 3. The carrier lifetime determined by this global fit is τ_r = 218 ± 3 ps. Note that a fit of the Δx = 0 transient in Figure 5B to a single-exponential function yields a carrier lifetime of 107 ps, which is nearly 2-fold shorter than the value determined from the fit to eq 3. This result highlights the substantial underestimation of the carrier lifetime that can result from an analysis that does not properly account for carrier diffusion.

Electron−Hole Recombination Kinetics. Ideally, SSPP kinetics like those in Figure 5 would be collected for each NW, allowing a global fit of carrier diffusion and recombination dynamics, but current experimental limitations make this impractical. However, with knowledge of pump and probe spot sizes, a single fit to the spatially overlapped decay (Δx = 0) using eq 3 should recover the correct carrier lifetime. To test this hypothesis, we fit the Δx = 0 kinetic trace in Figure 5B using eq 3 and recovered a lifetime of τ_S = 217 ± 6 ps. This value compares favorably with the lifetime determined by the global fit to the set of spatially separated kinetic traces. Similar agreement between the lifetimes obtained by the two methods was found across multiple data sets, indicating that eq 3 can disentangle the recombination and diffusion contributions and enable extraction of the carrier recombination lifetime from spatially overlapped kinetic traces.

The distribution of lifetimes in the NW samples was determined by fitting eq 3 to each of the 25−50 SOPP transient kinetic traces collected for the four samples (light blue curves in Figure 3). The distribution is plotted with green bars (which indicate the centers of 35 ps wide bins) in Figure 6, and a fit to the histogram with a Weibull model is shown as a solid black curve.

Because of their high surface to volume ratio, electron−hole recombination in NWs is expected to be dominated by surface traps.[7,9,18] The increase in average lifetime that is observed with increasing NW diameter is consistent with surface recombination as the primary decay mechanism. In this limit, bulk

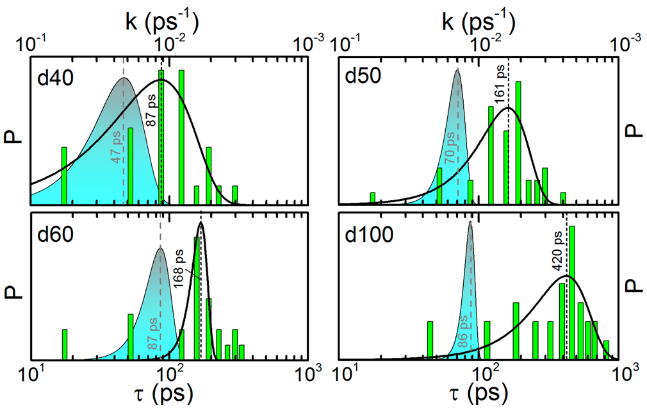

Figure 6. Comparison of lifetime distributions determined by pump−probe microscopy (green bars) and TA (blue shaded curve) for samples (A) d40, (B) d50, (C) d60, and (D) d100. The microscopy distribution is generated by fitting eq 3 to each of the spatially overlapped transients shown in Figure 3 to extract a carrier lifetime. The bin width is 35 ps for the lifetime histogram. The black solid line shows the fit of the microscopy data to a Weibull distribution. The TA lifetime distribution is determined by the SEM-measured diameters together with the spectroscopically measured surface recombination velocity. For each sample, the lifetime with maximum probability from TA and microscopy is noted next to each distribution.

recombination processes are negligible, and the carrier lifetime scales linearly with the NW diameter (d):

$$\tau_r = \frac{d}{4S} \tag{4}$$

where S is the surface recombination velocity, which reflects surface quality.[27−29] The values of S calculated from the average diameters and lifetimes measured for each sample (Table 1) are

Table 1. Summary of Surface Recombination Velocities Determined by Microscopy and TA

			S (10^4 cm/s)	
sample ID	d (nm)	lifetime, τ_r (ps)	microscopy[b]	TA[a]
d40	38	87	1.09 (0.08)	2.05 (0.09)
d50	52	161	0.81 (0.05)	1.87 (0.07)
d60	59	168	0.88 (0.02)	1.71 (0.06)
d100	101	420	0.60 (0.03)	2.96 (0.22)

[a]Errors shown in parentheses are 1 standard deviation in the fitted parameter, S. [b]Errors shown in parentheses are determined from 1 standard deviation uncertainty in the center of the fit (black solid lines, Figure 6) to the lifetimes.

comparable to recombination velocities observed in bulk materials[30] and single NWs.[6,18] Interestingly, the smaller diameter wires exhibit large recombination velocities, which could be a manifestation of the higher radius of curvature at their surface, which would presumably give rise to a greater defect density. While such a result is intriguing, variations from sample to sample make it difficult to draw definitive conclusions without additional experiments dedicated to resolving this effect.

Lifetimes obtained from pump−probe microscopy of individual NWs are ∼2−5 times longer than those obtained with TA (Figure 6).[18] The exact origin of this difference is unclear; however, because both experiments were performed on the same samples, it must arise from differences in the structures probed in the two studies. Unlike the microscopy experiment, which samples select regions of individual NWs, the TA experiment samples the full distribution of nanostructures as well as extraneous material, including side products, residual reactants, catalysts, etc., that still remain in the sample. Contributions from these undesired materials to the TA signal will skew the decay to shorter times, causing the recombination to appear faster than it actually is in NWs themselves. While SEM images of the samples suggest the volume of non-NW materials is too small to substantially influence the measured TA signal (Figure 1A), the larger absorption cross section of these materials (amorphous silicon has an extinction coefficient that is over an order of magnitude greater than that of crystalline silicon, for example) could cause them to contribute disproportionately to the total signal. While this makes it difficult to rule these structures out, it is unlikely that they are the primary source of the difference between the microscopy and TA results.

A second possibility stems from the fact that, although the TA and microscopy experiments were performed on the same sample, the two experiments probed different regions of the film. Femtosecond TA spectra were collected in areas of high NW density, where the optical absorption was greater. In these regions, the individual NWs are constrained in their orientation and adopt conformations with significant local curvature (Figure 1A). Microscopy measurements, on the other hand, were performed at the edge of the samples where there were fewer overlapping wires. While the lower density makes it possible to isolate individual NWs, it also results in wires with less curvature (Figure 1B). The difference between the microscopy and TA results may reflect this difference in NW secondary structure. In ZnO and CdS NWs, for example, curvature is known to strongly influence the electronic structure of the bulk lattice,[31,32] and the strain induced by bending could increase the defect density.[29] We anticipate that both would shorten the carrier lifetime, and could largely account for the differences observed between the two techniques. A quantitative analysis of the influence of secondary structure on the charge carrier lifetime is currently under way.

In summary, we have studied carrier recombination and diffusion in individual silicon NWs. SSPP microscopy was used to directly image carrier spatial evolution on ultrafast time scales. We have developed a model that allows carrier diffusion to be decoupled from the recombination dynamics in SOPP measurements, which should allow rapid characterization of many individual nanostructures within an ensemble. A survey of individual NWs grown with catalysts of various diameters shows a distribution of carrier recombination rates, which can be attributed to the distribution of NW diameters within each sample. On average, measurements of individual NWs exhibit longer lifetimes than ensemble measurements of the same NWs, highlighting the important role sample heterogeneity plays in determining the spectroscopic observables and functionality of nanostructures.

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Notes

The authors declare no competing financial interest.

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